

Vegetarian diet improves insulin resistance and oxidative stress markers more than conventional diet in subjects with Type 2 diabetes

Aims The aim of this study was to compare the effects of calorie-restricted vegetarian and conventional diabetic diets alone and in combination with exercise on insulin resistance, visceral fat and oxidative stress markers in subjects with Type 2 diabetes. **Methods** A 24-week, randomized, open, parallel design was used. Seventy-four patients with Type 2 diabetes were randomly assigned to either the experimental group (n = 37), which received a vegetarian diet, or the control group (n = 37), which received a conventional diabetic diet. Both diets were isocaloric, calorie restricted (-500 kcal/day). All meals during the study were provided. The second 12 weeks of the diet were combined with aerobic exercise. Participants were examined at baseline, 12 weeks and 24 weeks. Primary outcomes were: insulin sensitivity measured by hyperinsulinaemic isoglycaemic clamp; volume of visceral and subcutaneous fat measured by magnetic resonance imaging; and oxidative stress measured by thiobarbituric acid reactive substances. Analyses were by intention to treat. **Results** Forty-three per cent of participants in the experimental group and 5% of participants in the control group reduced diabetes medication ($P < 0.001$). Body weight decreased more in the experimental group than in the control group [-6.2 kg (95% CI -6.6 to -5.3) vs. -3.2 kg (95% CI -3.7 to -2.5); interaction group $\times$ time $P = 0.001$]. An increase in insulin sensitivity was significantly greater in the experimental group than in the control group [30% (95% CI 24.5–39) vs. 20% (95% CI 14–25), $P = 0.04$]. A reduction in both visceral and subcutaneous fat was greater in the experimental group than in the control group ($P = 0.007$ and $P = 0.02$, respectively). Plasma adiponectin increased ($P = 0.02$) and leptin decreased ($P = 0.02$) in the experimental group, with no change in the control group. Vitamin C, superoxide dismutase and reduced glutathione increased in the experimental group ($P = 0.002$, $P < 0.001$ and $P = 0.02$, respectively). Differences between groups were greater after the addition of exercise training. Changes in insulin sensitivity and enzymatic oxidative stress markers correlated with changes in visceral fat. **Conclusions** A calorie-restricted vegetarian diet had greater capacity to improve insulin sensitivity compared with a conventional diabetic diet over 24 weeks. The greater loss of visceral fat and improvements in plasma concentrations of adipokines and oxidative stress markers with this diet may be responsible for the reduction of insulin resistance. The addition of exercise training further augmented the improved outcomes with the vegetarian diet.